Parardha Dhar

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EDUCATION

VIT Bhopal University *Sep 2024 – 2028 | Bhopal, India*
Bachelor of Technology in Computer Science and Engineering **| GPA: 8.68 / 10.0** *Don Bosco School, Darjeeling — Graduated May 2024*

TECHNICAL SKILLS

Languages: Python, Java, C++, C#, JavaScript, SQL, CUDA
ML & Artificial Intelligence: PyTorch, TensorFlow, scikit-learn, OpenCV, HuggingFace, LangChain, Retrieval-Augmented Generation (RAG), Machine Learning
Web & Full-Stack Development: Next.js, React, Node.js, REST API Design, Microservices, Mobile-Responsive Design, Vercel
Cloud & DevOps: Oracle Cloud Infrastructure (OCI), Docker, Kubernetes, CI/CD, GitHub Actions, Application Resiliency, Security Best Practices
Databases: Supabase, Firebase, SQL, Relational Databases, Data Warehousing Concepts
Software Engineering: Data Structures and Algorithms, Object-Oriented Programming, System Design, Agile Methodologies, Software Testing, Business Analysis, Distributed Systems, Big Data Processing

EXPERIENCE

Founder & Lead Systems Engineer — Fractured Hours — Game Engine *Aug 2024 – Present*

- Architected and led end-to-end development of a large-scale, production-grade game engine with a cross-functional team of 8 engineers and artists, managing stakeholders and driving all system design decisions
- Engineered a spatial data structure overhaul that reduced large-scene traversal overhead by an estimated 40%, delivering measurably improved performance and application resiliency at runtime
- Designed, coded, and tested a data-driven event system using object-oriented programming and agile development practices that fully decoupled game logic from the engine core, improving scalability and software maintainability
- Implemented reinforcement learning and artificial intelligence experiments for NPC behaviour directly inside the engine, applying industry best practices to shorten the loop between ML research and live gameplay

Machine Learning & MLOps Engineer — Self-Directed Research *2024 – Present*

- Developed and deployed end-to-end machine learning pipelines in Python and Java-compatible frameworks, taking models from raw data ingestion through software testing to containerised cloud REST API inference endpoints
- Implemented CI/CD workflows, automated software testing, versioned datasets, and experiment tracking to ensure full reproducibility — applying agile methodologies and software improvement practices across all projects
- Optimised inference latency and application resiliency through async serving and request batching via Docker and Kubernetes, consistently meeting production performance targets across classification, computer vision, and generative AI tasks

PROJECTS

VRGC Community Hub — Full-Stack Web Platform · VR & Gaming Club, VIT Bhopal *2025*
Next.js, React, Supabase, Firebase, Vercel, REST APIs, CI/CD, Cloud Deployment | vrgc.vercel.app

- Architected and delivered a large-scale, resilient full-stack platform end-to-end, handling all design, coding, testing, and cloud deployment while managing live operations during a high-traffic production event
- Engineered the system to sustain 1,800+ unique visitors, 8,300+ page views, and 1,000+ clicks per second without bottlenecking — demonstrating application resiliency and scalable cloud architecture under real-world load
- Optimised live stream delivery for 800+ concurrent viewers using chunked transport and lazy loading, achieving a 96/100 Real Experience Score (First Contentful Paint 1.32s, Largest Contentful Paint 1.48s) under peak traffic

Real-Time Indian Sign Language Recognition System *2024*
Python, OpenCV, MediaPipe, TensorFlow, gTTS, REST API

- Developed a real-time artificial intelligence pipeline converting Indian Sign Language webcam input into text and multilingual speech — a technology solution built for social good, enabling communication access for the hearing-impaired community
- Implemented MediaPipe hand-landmark detection with a custom-trained TensorFlow classifier; optimised for low-resource and mobile hardware to enable real-world deployment outside a lab setting, with end-to-end latency under 100ms per frame

PhishGuard — Real-Time Cybersecurity Threat Detection System *2024*
Python, scikit-learn, Flask, Docker, REST API | github.com/parardhadhar/phishguard

- Engineered a machine learning-based URL classification system applying cybersecurity and security best practices to detect phishing threats in real time; containerised with Docker and deployed behind a REST API for seamless integration into enterprise security pipelines
- Applied business analysis thinking to prioritise low false-positive rate over recall — incorrect security flags damage user trust and business operations more than occasional misses, a deliberate product and software design decision

Document Insight Engine — Retrieval-Augmented Generation QA System *2024*

Python, LangChain, HuggingFace, FAISS, Big Data Processing | huggingface.co/spaces/parardhadhar/Parardha-Insight-Engine

- Built a multi-document question-answering system using Retrieval-Augmented Generation (RAG) and big data processing techniques; implemented FAISS vector store with chunk-level embedding to handle large document sets with fast, consistent query latency
- Deployed as a live public cloud application on Hugging Face Spaces — designed for accessibility without setup, making advanced AI available as a practical tool rather than a demo

Krishi Sahayak — AI-Powered Smart Farming Platform *2024*

Python, TensorFlow, Flask, REST API, Web Dashboard | github.com/parardhadhar/Krishi-Sahayak

- Developed a real-world technology solution for social good: a CNN-based crop disease classifier with a recommendation engine for fertilizer and harvest timing, delivered through a mobile-responsive web dashboard for smallholder farmers without agronomist access

CERTIFICATIONS

Oracle Cloud Infrastructure 2025 — Generative AI Professional (Oracle)
AI/ML for Geo Data Analysis (ISRO — Indian Space Research Organisation)
Supervised Machine Learning: Regression & Classification (DeepLearning.AI)
Machine Learning with Python (IBM)
Introduction to Security Principles in Cloud Computing (Google Cloud)
Competitive Programming: 100+ problems solved on LeetCode, Codeforces, GeeksForGeeks, and HackerRank — strong in Data Structures and Algorithms, dynamic programming, graphs, recursion, and greedy algorithms. Actively growing.

RELEVANT COURSEWORK

Data Structures and Algorithms · Operating Systems · Artificial Intelligence · Machine Learning · Computer Graphics · Database Management Systems · Unix Tools and Scripting · Real-Time Systems · Numerical Computing · Software Engineering